Dev Mehta

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AI Systems Engineer · Multi-Agent Systems · Biomedical AI · Cloud-Native Engineering · Causal ML

Education

SRM Institute of Science and Technology, Vadapalani — Chennai, India
*B.Tech, Computer Science and Engineering — CGPA: **9.21 / 10** — **Rank 1 – Overall Proficiency*** — *2023 – 2027*

Union Christian Public School (CBSE) — Chennai, India
*Class XII: **88.8%** Class X: **96.8%*** — *2021, 2023*

Experience

Jayaram Infotech — Remote
AI/ML Developer Intern — *Jan 2026 – Jun 2026*

- Contributed to **AksharabyasaAI**, an AI-powered EdTech platform for handwriting assessment and personalized learning.
- Engineered computer vision and OCR pipelines for handwritten character recognition using deep learning models.
- Developed evaluation systems analyzing stroke patterns, letter formation, and writing consistency for automated feedback generation.
- Integrated production-ready AI modules into a full-stack platform, collaborating across frontend and backend teams.

Projects

NEXUS – Multi-Agent Strategic Intelligence Platform — *Agentic AI*
Python, FastAPI, LangGraph, Vector Databases, Redis, React

- Orchestrated **5 specialized AI agents** (Researcher, Architect, Strategist, Critic, Memory) via a centralized planning and execution framework.
- Implemented **3-stage adversarial critique workflows** enabling iterative assumption-challenging and decision-ready synthesis across 4 strategic dimensions.
- Designed persistent memory mechanisms for cross-session knowledge retention, organizational learning, and strategic context reuse.

SELENE – Ambient Cognitive Operating Layer — *Distributed AI / Cloud*
Python, FastAPI, Kafka, Redis, Kubernetes, React

- Architected a cognitive operating layer with **5 independent subsystems** governing planning, execution, trust, memory, and reflection.
- Established **3-layer trust enforcement** via policy validation, hybrid critic evaluation, and permission-scoped execution controls.
- Structured autonomous workflows through deterministic DAG execution and lease-bound worker orchestration for auditable AI actions.

ATLAS – Causal Digital Twin for Single-Cell State Simulation — *Computational Biology*
Python, PyTorch Geometric, Scanpy, NetworkX, Causal ML

- Developed a causal digital twin framework integrating Graph Neural Networks, temporal simulation, and uncertainty-aware reasoning for single-cell state prediction.
- Validated model generalization across 3 benchmark biological datasets (Norman, Dixit, and Replogle) and incorporated TRRUST gene regulatory networks for causal modeling.
- Designed an active-learning experiment prioritization engine combining uncertainty estimation, diversity scoring, and intervention recommendation.
- Published research findings in **Wiley ICISD 2026 Proceedings**.

BioGPT-DDI – Graph-Augmented Drug Interaction Intelligence System — *Biomedical AI / NLP*
Python, PyTorch, HuggingFace Transformers, BioGPT, FastAPI, React

- Developed a graph-augmented biomedical NLP framework for drug-drug interaction prediction across 4 clinical interaction classes using BioBERT and BioGPT.
- Fine-tuned transformer-based classification pipelines on the DDI-2013 benchmark corpus, leveraging focal-loss optimization to address severe class imbalance.
- Achieved an F1-score of 86.64% for interaction classification while generating clinician- and patient-oriented explainable reports through BioGPT.
- Published findings in **IJIREEICE 2025** and deployed the system through FastAPI-based inference services and React interfaces.

Publications

BioGPT-DI: A Graph-Augmented LLM Framework for Drug Interaction Prediction and Explainable Clinical Reports – *IJIREEICE, 2025*

ATLAS: A Causal Digital Twin Framework for Single-Cell State Simulation – *Wiley ICISD 2026 Proceedings*

CEREP: A Graph-Constrained Explainable Reasoning Engine for Multi-Omics Precision Oncology – *IJARCCE*

Leadership & Achievements

Top 1000 Builders shortlisted from 10,000+ applicants – **OpenAI** × **Outskill AI Builders Program** (2026)

President, Design and Innovation (DI) Club — **Google Gemini Student Ambassador 2026** — Aperture 3.0 Open Source Contributor

Top 50 Teams – Web of Innovation Hackathon, TechVerse (MedIntel AI: BioGPT-based clinical decision support system, Mar 2026)

Rank 1 – Overall Proficiency Award, B.Tech CSE, SRM IST Vadapalani (Academic Excellence, Annual Day 2025–26)

1st Place – DesignVerse UI/UX Challenge (Jul 2025) — **3rd Place** – AceInt Platform Redesign Challenge (Aug 2025)

Technical Skills

Languages: Python, Java, C++, JavaScript

AI & Machine Learning: PyTorch, HuggingFace Transformers, LangGraph, Agentic AI, LLM Systems, Graph Neural Networks (GNNs), Causal Inference, Explainable AI (XAI), Time-Series ML

Distributed Systems & Cloud: Kafka, Redis, Kubernetes, Docker, Ray, Microservices, Event-Driven Architectures

Backend & APIs: FastAPI, REST APIs, Vector Databases

Frontend: React.js, HTML, CSS

Bioinformatics & Scientific Computing: PyTorch Geometric, Scanpy, NetworkX, Knowledge Graphs, Multi-Omics Analysis, Computational Biology

Tools: Git, GitHub, MLflow, Optuna, Figma